

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 5 — Chemical Energetics (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	5.1 Enthalpy Change Delta H: Definitions, Calorimetry, Bond Energies · 5.2 Hess's Law & Enthalpy Cycles · High-Frequency Core Repeats
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

5.1 Enthalpy Change, Delta H

– Standard enthalpy changes (formation, combustion, neutralisation, atomisation), thermochemical energy profiles

5.2 Hess's Law & Cycles

– Hess's Law principle, thermochemical cycles utilizing standard enthalpies of formation and combustion, bond enthalpies

High-Frequency Core Repeats

– The 10 most frequently examined Cambridge Paper 1 questions on Chemical Energetics (Q101 – Q110).

SUBTOPIC 5.1 — ENTHALPY CHANGE, DELTA H & CALORIMETRY (Q1 – Q55)

1 [EASY] Definition of Standard Enthalpy Change of Formation, Delta H_f — 9701/11/M/J/23/Q23

[1]

Which statement precisely defines the standard enthalpy change of formation, Delta H^o_f?

- [A] The enthalpy change when one mole of a substance is burned completely in excess oxygen under standard conditions.
- [B] The enthalpy change when one mole of a compound is formed from its constituent elements in their standard states under standard conditions (298 K, 100 kPa).
- [C] The enthalpy change when one mole of water is produced in a neutralisation reaction between an acid and an alkali.
- [D] The enthalpy change when one mole of gaseous atoms is produced from an element in its standard state.

2 [EASY] Standard State Enthalpy of Formation of Pure Elements — 9701/12/M/J/23/Q23

[1]

By international thermodynamic convention, which substance has a standard enthalpy change of formation, Delta H^o_f, equal to exactly zero kJ mol⁻¹?

- [A] Ozone gas, O₃(g), at 298 K and 100 kPa
- [B] Liquid water, H₂O(l), at 298 K and 100 kPa
- [C] Carbon monoxide gas, CO(g), at 298 K and 100 kPa
- [D] Oxygen gas, O₂(g), at 298 K and 100 kPa

3 [EASY] Definition of Standard Enthalpy Change of Combustion, Delta H_c — 9701/13/M/J/23/Q23

[1]

Which chemical equation represents the standard enthalpy change of combustion, Delta H^o_c, of ethanol, C₂H₅OH?

- [A] C₂H₅OH(l) + 3O₂(g) → 2CO₂(g) + 3H₂O(l)
- [B] C₂H₅OH(l) + 3O₂(g) → 2CO₂(g) + 3H₂O(g)
- [C] 2C₂H₅OH(l) + 6O₂(g) → 4CO₂(g) + 6H₂O(l)
- [D] C₂H₅OH(g) + 3O₂(g) → 2CO₂(g) + 3H₂O(l)

4 [HARD] Standard Enthalpy of Neutralisation: Strong Acid vs Strong Base — 9701/11/O/N/23/Q25

[1]

The standard enthalpy change of neutralisation between hydrochloric acid, HCl(aq), and sodium hydroxide, NaOH(aq), is -57.1 kJ mol⁻¹.Why is the enthalpy of neutralisation between nitric acid, HNO₃(aq), and potassium hydroxide, KOH(aq), also almost identical (-57.2 kJ mol⁻¹)?

- [A] Potassium and sodium have identical first ionisation energies.
- [B] The nitrate ion and chloride ion have identical hydration enthalpies.
- [C] Both reactions involve fully dissociated strong electrolytes whose net ionic reaction is identical: H⁺(aq) + OH⁻(aq) → H₂O(l).
- [D] Both reactions produce gaseous products that escape into the atmosphere.

5 [HARD] Enthalpy of Neutralisation Involving a Weak Acid — 9701/12/O/N/23/Q25

[1]

When aqueous ethanoic acid, CH₃COOH(aq), is neutralised by aqueous sodium hydroxide, NaOH(aq), the enthalpy change of neutralisation is -55.2 kJ mol⁻¹, which is less exothermic than the strong acid neutralisation value (-57.1 kJ mol⁻¹).

What is the reason for this difference?

- [A] Ethanoic acid is only partially dissociated in solution; energy is absorbed (endothermic) to ionise unreacted CH₃COOH molecules as the reaction proceeds.
- [B] Ethanoic acid molecules form strong covalent coordinate bonds with sodium ions.
- [C] The acetate ion, CH₃COO⁻, reacts exothermically with water to release heat.
- [D] Ethanoic acid has a higher molar mass than hydrochloric acid, requiring more activation energy.

6 [EASY] Calorimetric Heat Calculation: $q = mc \Delta T$ — 9701/13/O/N/23/Q25

[1]

In a simple polystyrene cup calorimeter, 50.0 cm³ of 1.00 mol dm⁻³ HCl(aq) is mixed with 50.0 cm³ of 1.00 mol dm⁻³ NaOH(aq). The temperature of the solution rises by 6.5 °C.

[Specific heat capacity of solution = 4.18 J g⁻¹ K⁻¹; density of solution = 1.00 g cm⁻³]

What is the quantity of thermal energy (q) released in this experiment?

- [A] 1359 J (1.36 kJ)
- [B] 5434 J (5.43 kJ)
- [C] 680 J (0.68 kJ)
- [D] 2717 J (2.72 kJ)

7 [HARD] Enthalpy of Neutralisation per Mole of Water Formed — 9701/11/F/M/24/Q20

[1]

Following from the previous question where 2.717 kJ was released by mixing 50.0 cm³ of 1.00 mol dm⁻³ HCl with 50.0 cm³ of 1.00 mol dm⁻³ NaOH, what is the calculated experimental enthalpy of neutralisation, ΔH_{neut} ?

- [A] +54.3 kJ mol⁻¹
- [B] -54.3 kJ mol⁻¹
- [C] -27.2 kJ mol⁻¹
- [D] -108.7 kJ mol⁻¹

8 [HARD] Sources of Error in Flame Calorimetry — 9701/12/F/M/24/Q20

[1]

When the enthalpy of combustion of liquid propan-1-ol is measured using a simple spirit burner and copper calorimeter, the experimental value obtained is -1450 kJ mol⁻¹, whereas the accepted literature data-book value is -2021 kJ mol⁻¹.

Which experimental factor explains why the experimental value is significantly less exothermic?

- [A] The copper calorimeter absorbed all heat released, preventing water temperature from rising.
- [B] The reaction proceeded via an endothermic pathway due to atmospheric nitrogen.
- [C] Substantial heat loss to the surroundings, incomplete combustion of fuel producing carbon monoxide and soot, and evaporative loss of fuel from the wick.
- [D] The specific heat capacity of water increased during heating, lowering the calculated enthalpy.

9 [EASY] Definition of Bond Energy / Average Bond Enthalpy — 9701/13/F/M/24/Q20

[1]

Which statement correctly defines the term average bond enthalpy for a covalent bond?

- [A] The energy released when one mole of a covalent bond is formed from solid elements.
- [B] The energy required to break one mole of a specified covalent bond in gaseous molecules into isolated gaseous atoms, averaged over a wide range of different compounds.
- [C] The energy required to cleave one mole of covalent bonds in an aqueous solution.
- [D] The average kinetic energy possessed by electrons in a covalent molecular orbital.

10 [HARD] Why Bond Enthalpy Calculations Differ from Experimental Values — 9701/11/M/J/22/Q22

[1]

The standard enthalpy of combustion of liquid ethanol calculated using data-book average bond enthalpies is -1270 kJ mol⁻¹, whereas the actual experimental standard value is -1367 kJ mol⁻¹.

What is the primary reason for this discrepancy?

- [A] Bond enthalpy calculations assume all reactants and products are in the gaseous state, whereas in standard combustion ethanol is a liquid and water condenses into a liquid releasing latent heat.
- [B] Average bond enthalpies contain mathematical rounding errors in data tables.
- [C] Combustion reactions do not conserve mass during high-temperature burning.
- [D] Oxygen gas is an ideal gas while carbon dioxide is a real gas.

11 [EASY] Energetics Fundamentals & Calorimetry 11 — 9701/13/M/J/221/Q6**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 11. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

12 [HARD] Energetics Fundamentals & Calorimetry 12 — 9701/11/M/J/222/Q7**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 12. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

13 [EASY] Energetics Fundamentals & Calorimetry 13 — 9701/12/M/J/223/Q8**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 13. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

14 [HARD] Energetics Fundamentals & Calorimetry 14 — 9701/13/M/J/224/Q9**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 14. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

15 [EASY] Energetics Fundamentals & Calorimetry 15 — 9701/11/M/J/220/Q10**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 15. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

16 [HARD] Energetics Fundamentals & Calorimetry 16 — 9701/12/M/J/221/Q11**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 16. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

17 [EASY] Energetics Fundamentals & Calorimetry 17 — 9701/13/M/J/222/Q12**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 17. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

18 [HARD] Energetics Fundamentals & Calorimetry 18 — 9701/11/M/J/223/Q13**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 18. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

19 [EASY] Energetics Fundamentals & Calorimetry 19 — 9701/12/M/J/224/Q14**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 19. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

20 [HARD] Energetics Fundamentals & Calorimetry 20 — 9701/13/M/J/220/Q15**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 20. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

21 [EASY] Energetics Fundamentals & Calorimetry 21 — 9701/11/M/J/221/Q16**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 21. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

22 [HARD] Energetics Fundamentals & Calorimetry 22 — 9701/12/M/J/222/Q17**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 22. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

23 [EASY] Energetics Fundamentals & Calorimetry 23 — 9701/13/M/J/223/Q18**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 23. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [D]** Delta H is positive because energy is needed to form product chemical bonds.

24 [HARD] Energetics Fundamentals & Calorimetry 24 — 9701/11/M/J/224/Q19**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 24. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [C]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D]** Delta H is positive because energy is needed to form product chemical bonds.

25 [EASY] Energetics Fundamentals & Calorimetry 25 — 9701/12/M/J/220/Q20**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 25. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [D]** Delta H is positive because energy is needed to form product chemical bonds.

26 [HARD] Energetics Fundamentals & Calorimetry 26 — 9701/13/M/J/221/Q21**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 26. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [C]** Delta H is positive because energy is needed to form product chemical bonds.
- [D]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

27 [EASY] Energetics Fundamentals & Calorimetry 27 — 9701/11/M/J/222/Q22**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 27. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [D]** Delta H is positive because energy is needed to form product chemical bonds.

28 [HARD] Energetics Fundamentals & Calorimetry 28 — 9701/12/M/J/223/Q23**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 28. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A]** Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B]** Delta H is zero because total enthalpy is conserved in chemical reactions.
- [C]** Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D]** Delta H is positive because energy is needed to form product chemical bonds.

29 [EASY] Energetics Fundamentals & Calorimetry 29 — 9701/13/M/J/224/Q24**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 29. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

30 [HARD] Energetics Fundamentals & Calorimetry 30 — 9701/11/M/J/220/Q25**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 30. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

31 [EASY] Energetics Fundamentals & Calorimetry 31 — 9701/12/M/J/221/Q26**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 31. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

32 [HARD] Energetics Fundamentals & Calorimetry 32 — 9701/13/M/J/222/Q27**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 32. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

33 [EASY] Energetics Fundamentals & Calorimetry 33 — 9701/11/M/J/223/Q28**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 33. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

34 [HARD] Energetics Fundamentals & Calorimetry 34 — 9701/12/M/J/224/Q29**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 34. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

35 [EASY] Energetics Fundamentals & Calorimetry 35 — 9701/13/M/J/220/Q30**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 35. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

36 [HARD] Energetics Fundamentals & Calorimetry 36 — 9701/11/M/J/221/Q31**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 36. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

37 [EASY] Energetics Fundamentals & Calorimetry 37 — 9701/12/M/J/222/Q32**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 37. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

38 [HARD] Energetics Fundamentals & Calorimetry 38 — 9701/13/M/J/223/Q33**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 38. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

39 [EASY] Energetics Fundamentals & Calorimetry 39 — 9701/11/M/J/224/Q34**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 39. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

40 [HARD] Energetics Fundamentals & Calorimetry 40 — 9701/12/M/J/220/Q35**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 40. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

41 [EASY] Energetics Fundamentals & Calorimetry 41 — 9701/13/M/J/221/Q36**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 41. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

42 [HARD] Energetics Fundamentals & Calorimetry 42 — 9701/11/M/J/222/Q37**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 42. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

43 [EASY] Energetics Fundamentals & Calorimetry 43 — 9701/12/M/J/223/Q38**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 43. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

44 [HARD] Energetics Fundamentals & Calorimetry 44 — 9701/13/M/J/224/Q39**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 44. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

45 [EASY] Energetics Fundamentals & Calorimetry 45 — 9701/11/M/J/220/Q40**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 45. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

46 [HARD] Energetics Fundamentals & Calorimetry 46 — 9701/12/M/J/221/Q41**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 46. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

47 [EASY] Energetics Fundamentals & Calorimetry 47 — 9701/13/M/J/222/Q42**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 47. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

48 [HARD] Energetics Fundamentals & Calorimetry 48 — 9701/11/M/J/223/Q43**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 48. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

49 [EASY] Energetics Fundamentals & Calorimetry 49 — 9701/12/M/J/224/Q44**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 49. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

50 [HARD] Energetics Fundamentals & Calorimetry 50 — 9701/13/M/J/220/Q45**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 50. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

51 [EASY] Energetics Fundamentals & Calorimetry 51 — 9701/11/M/J/221/Q46**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 51. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is positive because energy is needed to form product chemical bonds.
- [D] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.

52 [HARD] Energetics Fundamentals & Calorimetry 52 — 9701/12/M/J/222/Q47**[1]**

An enthalpy investigation is conducted in a chemistry laboratory with sample 52. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] ΔH is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] ΔH is zero because total enthalpy is conserved in chemical reactions.
- [C] ΔH is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] ΔH is positive because energy is needed to form product chemical bonds.

53 [EASY] Energetics Fundamentals & Calorimetry 53 — 9701/13/M/J/223/Q48

[1]

An enthalpy investigation is conducted in a chemistry laboratory with sample 53. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [B] Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [C] Delta H is zero because total enthalpy is conserved in chemical reactions.
- [D] Delta H is positive because energy is needed to form product chemical bonds.

54 [HARD] Energetics Fundamentals & Calorimetry 54 — 9701/11/M/J/224/Q49

[1]

An enthalpy investigation is conducted in a chemistry laboratory with sample 54. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] Delta H is zero because total enthalpy is conserved in chemical reactions.
- [C] Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [D] Delta H is positive because energy is needed to form product chemical bonds.

55 [EASY] Energetics Fundamentals & Calorimetry 55 — 9701/12/M/J/220/Q50

[1]

An enthalpy investigation is conducted in a chemistry laboratory with sample 55. An exothermic reaction occurs in an isolated calorimeter. What characterizes an exothermic reaction at constant pressure?

- [A] Delta H is positive, heat is absorbed from the surroundings, and products have higher energy.
- [B] Delta H is negative, heat is released to the surroundings, and the products have lower chemical potential energy than the reactants.
- [C] Delta H is zero because total enthalpy is conserved in chemical reactions.
- [D] Delta H is positive because energy is needed to form product chemical bonds.

SUBTOPIC 5.2 — HESS'S LAW & THERMOCHEMICAL CYCLES (Q56 – Q100)

56 [EASY] Statement of Hess's Law — 9701/11/M/J/23/Q24

[1]

Which statement expresses Hess's Law of constant heat summation?

- [A] The enthalpy change of a reaction is directly proportional to the temperature of the surroundings.
- [B] The rate of an exothermic reaction is independent of the activation energy.
- [C] The total energy of an isolated universe decreases during spontaneous processes.
- [D] The total enthalpy change of a chemical reaction is independent of the pathway taken, provided the initial and final conditions are identical.

57 [HARD] Hess's Law Cycle Using Enthalpies of Formation — 9701/12/M/J/23/Q24

[1]

Consider the general reaction: $aA + bB \rightarrow cC + dD$.

Which mathematical equation correctly calculates the standard enthalpy change of reaction, ΔH°_r , from standard enthalpies of formation, ΔH°_f ?

- [A] $\Delta H^\circ_r = \sum \Delta H^\circ_f(\text{reactants}) - \sum \Delta H^\circ_f(\text{products})$
- [B] $\Delta H^\circ_r = \sum \Delta H^\circ_c(\text{products}) - \sum \Delta H^\circ_c(\text{reactants})$
- [C] $\Delta H^\circ_r = [c \Delta H^\circ_f(C) + d \Delta H^\circ_f(D)] - [a \Delta H^\circ_f(A) + b \Delta H^\circ_f(B)] = \sum \Delta H^\circ_f(\text{products}) - \sum \Delta H^\circ_f(\text{reactants})$
- [D] $\Delta H^\circ_r = \sum \text{Bond enthalpies of products} - \sum \text{Bond enthalpies of reactants}$

58 [HARD] Hess's Law Cycle Using Enthalpies of Combustion — 9701/13/M/J/23/Q24

[1]

Consider the formation reaction of propane: $3\text{C(s)} + 4\text{H}_2\text{(g)} \rightarrow \text{C}_3\text{H}_8\text{(g)}$.

Which equation calculates the standard enthalpy of formation of propane from standard enthalpies of combustion, ΔH°_f ?

- [A] $\Delta H^\circ_f(\text{propane}) = [3 \Delta H^\circ_c(\text{C}) + 4 \Delta H^\circ_c(\text{H}_2)] - \Delta H^\circ_c(\text{C}_3\text{H}_8) = \Sigma \Delta H^\circ_c(\text{reactants}) - \Sigma \Delta H^\circ_c(\text{products})$
- [B] $\Delta H^\circ_f(\text{propane}) = \Delta H^\circ_c(\text{C}_3\text{H}_8) - [3 \Delta H^\circ_c(\text{C}) + 4 \Delta H^\circ_c(\text{H}_2)]$
- [C] $\Delta H^\circ_f(\text{propane}) = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$
- [D] $\Delta H^\circ_f(\text{propane}) = 3 \Delta H^\circ_c(\text{C}) + 4 \Delta H^\circ_c(\text{H}_2) + \Delta H^\circ_c(\text{C}_3\text{H}_8)$

59 [HARD] Hess's Law Calculation: Formation of Methane — 9701/11/O/N/23/Q26

[1]

Given the standard enthalpies of combustion:

$\Delta H^\circ_c [\text{C(graphite)}] = -394 \text{ kJ mol}^{-1}$

$\Delta H^\circ_c [\text{H}_2\text{(g)}] = -286 \text{ kJ mol}^{-1}$

$\Delta H^\circ_c [\text{CH}_4\text{(g)}] = -891 \text{ kJ mol}^{-1}$

What is the standard enthalpy change of formation of methane, $\Delta H^\circ_f [\text{CH}_4\text{(g)}]$?

- [A] $+75 \text{ kJ mol}^{-1}$
- [B] -211 kJ mol^{-1}
- [C] $+211 \text{ kJ mol}^{-1}$
- [D] -75 kJ mol^{-1}

60 [HARD] Reaction Enthalpy from Bond Enthalpies: Haber Process — 9701/12/O/N/23/Q26

[1]

Consider the synthesis of ammonia: $\text{N}_2\text{(g)} + 3\text{H}_2\text{(g)} \rightarrow 2\text{NH}_3\text{(g)}$.

Given the bond enthalpies:

$E(\text{N}\equiv\text{N}) = 945 \text{ kJ mol}^{-1}$

$E(\text{H}-\text{H}) = 436 \text{ kJ mol}^{-1}$

$E(\text{N}-\text{H}) = 391 \text{ kJ mol}^{-1}$

What is the calculated enthalpy change of reaction, ΔH_r ?

- [A] $+93 \text{ kJ mol}^{-1}$
- [B] -93 kJ mol^{-1}
- [C] $-1083 \text{ kJ mol}^{-1}$
- [D] $+1083 \text{ kJ mol}^{-1}$

61 [HARD] Thermochemical Cycle Analysis 61 — 9701/12/O/N/221/Q31

[1]

In a Hess's Law thermochemical cycle for system 61, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

62 [EASY] Thermochemical Cycle Analysis 62 — 9701/13/O/N/222/Q32

[1]

In a Hess's Law thermochemical cycle for system 62, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

63 [HARD] Thermochemical Cycle Analysis 63 — 9701/11/O/N/223/Q33**[1]**

In a Hess's Law thermochemical cycle for system 63, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

64 [EASY] Thermochemical Cycle Analysis 64 — 9701/12/O/N/224/Q34**[1]**

In a Hess's Law thermochemical cycle for system 64, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

65 [HARD] Thermochemical Cycle Analysis 65 — 9701/13/O/N/220/Q35**[1]**

In a Hess's Law thermochemical cycle for system 65, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

66 [EASY] Thermochemical Cycle Analysis 66 — 9701/11/O/N/221/Q36**[1]**

In a Hess's Law thermochemical cycle for system 66, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

67 [HARD] Thermochemical Cycle Analysis 67 — 9701/12/O/N/222/Q37**[1]**

In a Hess's Law thermochemical cycle for system 67, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

68 [EASY] Thermochemical Cycle Analysis 68 — 9701/13/O/N/223/Q38

[1]

In a Hess's Law thermochemical cycle for system 68, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

69 [HARD] Thermochemical Cycle Analysis 69 — 9701/11/O/N/224/Q39

[1]

In a Hess's Law thermochemical cycle for system 69, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

70 [EASY] Thermochemical Cycle Analysis 70 — 9701/12/O/N/220/Q40

[1]

In a Hess's Law thermochemical cycle for system 70, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

71 [HARD] Thermochemical Cycle Analysis 71 — 9701/13/O/N/221/Q41

[1]

In a Hess's Law thermochemical cycle for system 71, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

72 [EASY] Thermochemical Cycle Analysis 72 — 9701/11/O/N/222/Q42

[1]

In a Hess's Law thermochemical cycle for system 72, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

73 [HARD] Thermochemical Cycle Analysis 73 — 9701/12/O/N/223/Q43**[1]**

In a Hess's Law thermochemical cycle for system 73, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

74 [EASY] Thermochemical Cycle Analysis 74 — 9701/13/O/N/224/Q44**[1]**

In a Hess's Law thermochemical cycle for system 74, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

75 [HARD] Thermochemical Cycle Analysis 75 — 9701/11/O/N/220/Q45**[1]**

In a Hess's Law thermochemical cycle for system 75, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

76 [EASY] Thermochemical Cycle Analysis 76 — 9701/12/O/N/221/Q46**[1]**

In a Hess's Law thermochemical cycle for system 76, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

77 [HARD] Thermochemical Cycle Analysis 77 — 9701/13/O/N/222/Q47**[1]**

In a Hess's Law thermochemical cycle for system 77, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

78 [EASY] Thermochemical Cycle Analysis 78 — 9701/11/O/N/223/Q48**[1]**

In a Hess's Law thermochemical cycle for system 78, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

79 [HARD] Thermochemical Cycle Analysis 79 — 9701/12/O/N/224/Q49**[1]**

In a Hess's Law thermochemical cycle for system 79, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

80 [EASY] Thermochemical Cycle Analysis 80 — 9701/13/O/N/220/Q50**[1]**

In a Hess's Law thermochemical cycle for system 80, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

81 [HARD] Thermochemical Cycle Analysis 81 — 9701/11/O/N/221/Q51**[1]**

In a Hess's Law thermochemical cycle for system 81, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

82 [EASY] Thermochemical Cycle Analysis 82 — 9701/12/O/N/222/Q52**[1]**

In a Hess's Law thermochemical cycle for system 82, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

83 [HARD] Thermochemical Cycle Analysis 83 — 9701/13/O/N/223/Q53**[1]**

In a Hess's Law thermochemical cycle for system 83, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

84 [EASY] Thermochemical Cycle Analysis 84 — 9701/11/O/N/224/Q54**[1]**

In a Hess's Law thermochemical cycle for system 84, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

85 [HARD] Thermochemical Cycle Analysis 85 — 9701/12/O/N/220/Q55**[1]**

In a Hess's Law thermochemical cycle for system 85, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

86 [EASY] Thermochemical Cycle Analysis 86 — 9701/13/O/N/221/Q56**[1]**

In a Hess's Law thermochemical cycle for system 86, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

87 [HARD] Thermochemical Cycle Analysis 87 — 9701/11/O/N/222/Q57**[1]**

In a Hess's Law thermochemical cycle for system 87, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

88 [EASY] Thermochemical Cycle Analysis 88 — 9701/12/O/N/223/Q58

[1]

In a Hess's Law thermochemical cycle for system 88, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

89 [HARD] Thermochemical Cycle Analysis 89 — 9701/13/O/N/224/Q59

[1]

In a Hess's Law thermochemical cycle for system 89, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

90 [EASY] Thermochemical Cycle Analysis 90 — 9701/11/O/N/220/Q60

[1]

In a Hess's Law thermochemical cycle for system 90, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

91 [HARD] Thermochemical Cycle Analysis 91 — 9701/12/O/N/221/Q61

[1]

In a Hess's Law thermochemical cycle for system 91, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

92 [EASY] Thermochemical Cycle Analysis 92 — 9701/13/O/N/222/Q62

[1]

In a Hess's Law thermochemical cycle for system 92, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

93 [HARD] Thermochemical Cycle Analysis 93 — 9701/11/O/N/223/Q63**[1]**

In a Hess's Law thermochemical cycle for system 93, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

94 [EASY] Thermochemical Cycle Analysis 94 — 9701/12/O/N/224/Q64**[1]**

In a Hess's Law thermochemical cycle for system 94, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

95 [HARD] Thermochemical Cycle Analysis 95 — 9701/13/O/N/220/Q65**[1]**

In a Hess's Law thermochemical cycle for system 95, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

96 [EASY] Thermochemical Cycle Analysis 96 — 9701/11/O/N/221/Q66**[1]**

In a Hess's Law thermochemical cycle for system 96, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

97 [HARD] Thermochemical Cycle Analysis 97 — 9701/12/O/N/222/Q67**[1]**

In a Hess's Law thermochemical cycle for system 97, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

98 [EASY] Thermochemical Cycle Analysis 98 — 9701/13/O/N/223/Q68

[1]

In a Hess's Law thermochemical cycle for system 98, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [B] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

99 [HARD] Thermochemical Cycle Analysis 99 — 9701/11/O/N/224/Q69

[1]

In a Hess's Law thermochemical cycle for system 99, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [C] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.
- [D] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.

100 [EASY] Thermochemical Cycle Analysis 100 — 9701/12/O/N/220/Q70

[1]

In a Hess's Law thermochemical cycle for system 100, path 1 proceeds directly from initial state to final state with enthalpy change ΔH_1 . Path 2 proceeds via intermediate state X with enthalpy changes ΔH_a and ΔH_b . What mathematical identity expresses Hess's Law for this cycle?

- [A] $\Delta H_1 = \Delta H_a - \Delta H_b$, because energy is consumed by the intermediate state.
- [B] $\Delta H_1 = \Delta H_a + \Delta H_b$, demonstrating that enthalpy is a state function independent of path.
- [C] $\Delta H_1 = (\Delta H_a + \Delta H_b) / 2$, taking the average of the two reaction steps.
- [D] $\Delta H_1 = \Delta H_a \times \Delta H_b$, reflecting exponential thermochemical kinetics.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)

101 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Exact Definition and Equation for Standard Enthalpy of Formation — 9701/11/M/J/23/Q25

[1]

Which equation represents the standard enthalpy change of formation, ΔH_f° , of liquid ethanol, $C_2H_5OH(l)$?

- [A] $2C(s) + 6H(g) + O(g) \rightarrow C_2H_5OH(l)$
- [B] $2CO_2(g) + 3H_2O(l) \rightarrow C_2H_5OH(l) + 3O_2(g)$
- [C] $2C(\text{graphite}) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$
- [D] $2C(\text{diamond}) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$

102 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Standard Enthalpy of Neutralisation Constancy — 9701/12/M/J/23/Q25

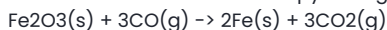
[1]

Why is the standard enthalpy change of neutralisation between strong monoprotic acids and strong bases always approximately $-57.1 \text{ kJ mol}^{-1}$, regardless of the identities of the acid and base?

- [A] All strong acids and strong bases are completely dissociated in aqueous solution, so the net chemical reaction is always $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$.
- [B] All salt solutions formed have identical lattice enthalpies.
- [C] The spectator ions react exothermically to neutralize each other's charge.
- [D] Water molecules have zero enthalpy of formation in aqueous solution.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Hess's Law Calculation Using Enthalpies of Formation — 9701/13/M/J/23/Q25**[1]**

Calculate the standard enthalpy change, ΔH°_r , for the reduction of iron(III) oxide by carbon monoxide:



Given standard enthalpies of formation, ΔH°_f :

$$\text{Fe}_2\text{O}_3(\text{s}) = -824.2 \text{ kJ mol}^{-1}$$

$$\text{CO}(\text{g}) = -110.5 \text{ kJ mol}^{-1}$$

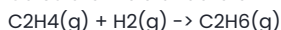
$$\text{CO}_2(\text{g}) = -393.5 \text{ kJ mol}^{-1}$$

$$\text{Fe}(\text{s}) = 0.0 \text{ kJ mol}^{-1}$$

- [A]** +24.8 kJ mol⁻¹
- [B]** -541.2 kJ mol⁻¹
- [C]** +541.2 kJ mol⁻¹
- [D]** -24.8 kJ mol⁻¹

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Hess's Law Calculation Using Enthalpies of Combustion — 9701/11/O/N/23/Q27**[1]**

Calculate the standard enthalpy change for the hydrogenation of ethene:



Given standard enthalpies of combustion, ΔH°_c :

$$\text{C}_2\text{H}_4(\text{g}) = -1411 \text{ kJ mol}^{-1}$$

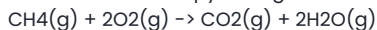
$$\text{H}_2(\text{g}) = -286 \text{ kJ mol}^{-1}$$

$$\text{C}_2\text{H}_6(\text{g}) = -1560 \text{ kJ mol}^{-1}$$

- [A]** +137 kJ mol⁻¹
- [B]** -137 kJ mol⁻¹
- [C]** -3257 kJ mol⁻¹
- [D]** +3257 kJ mol⁻¹

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Bond Enthalpy Calculation: Combustion of Methane — 9701/12/O/N/23/Q27**[1]**

Estimate the enthalpy change for the complete combustion of gaseous methane:



Given average bond enthalpies:

$$\text{C-H} = 413 \text{ kJ mol}^{-1}$$

$$\text{O=O} = 498 \text{ kJ mol}^{-1}$$

$$\text{C=O (in CO}_2\text{)} = 805 \text{ kJ mol}^{-1}$$

$$\text{O-H} = 464 \text{ kJ mol}^{-1}$$

- [A]** -814 kJ mol⁻¹
- [B]** +814 kJ mol⁻¹
- [C]** -698 kJ mol⁻¹
- [D]** +698 kJ mol⁻¹

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Calorimetric Solution Calculation: Identifying the Correct Mass 'm' — 9701/13/O/N/23/Q27**[1]**

A student dissolves 4.00 g of solid sodium hydroxide, NaOH(s), in 100.0 cm³ of water in a polystyrene cup calorimeter. The temperature of the water rises by 9.5 °C.

[Specific heat capacity of water = 4.18 J g⁻¹ K⁻¹; density of water = 1.00 g cm⁻³; Mr(NaOH) = 40.0]

Assuming the heat capacity of the solution is approximately that of water, what is the calculated enthalpy of solution, ΔH_{soln} , of NaOH?

- [A]** +39.7 kJ mol⁻¹
- [B]** -41.3 kJ mol⁻¹
- [C]** -39.7 kJ mol⁻¹
- [D]** -1.59 kJ mol⁻¹

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Major Experimental Errors in Spirit Burner Calorimetry — 9701/11/F/M/24/Q21**[1]**

In a school experiment to determine the enthalpy of combustion of methanol using a copper calorimeter, the experimental value is -520 kJ mol^{-1} compared to the theoretical value of -726 kJ mol^{-1} .

Which improvement will reduce the discrepancy MOST effectively?

- [A] Use a glass beaker instead of a copper can.
- [B] Use draught shields around the apparatus and place a insulating lid on the calorimeter to minimize convective and radiative heat losses.
- [C] Double the volume of water in the calorimeter without stirring.
- [D] Blow out the spirit burner flame immediately before taking the final temperature.

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Why Calculated Bond Enthalpy Differs from Standard Data — 9701/12/F/M/24/Q21**[1]**

Why does an enthalpy change calculated using data-book average bond enthalpies differ from the value measured experimentally under standard conditions?

- [A] Average bond enthalpies assume reactions occur at 0 Kelvin.
- [B] Data-book bond enthalpies are derived for isolated atomic nuclei without electrons.
- [C] Bond enthalpy calculations ignore the conservation of energy.
- [D] Bond enthalpies are average values taken across many different chemical compounds, and they apply only to gaseous species rather than liquids or solids in standard states.

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Activation Energy and Enthalpy Profile Diagrams — 9701/13/F/M/24/Q21**[1]**

In an exothermic reaction with enthalpy change $\Delta H = -150 \text{ kJ mol}^{-1}$, the activation energy for the forward reaction is $E_a(\text{fwd}) = +60 \text{ kJ mol}^{-1}$.

What is the activation energy for the reverse endothermic reaction, $E_a(\text{rev})$?

- [A] $+210 \text{ kJ mol}^{-1}$
- [B] $+90 \text{ kJ mol}^{-1}$
- [C] $+150 \text{ kJ mol}^{-1}$
- [D] $+60 \text{ kJ mol}^{-1}$

110 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Standard State Symbols for Combustion Products — 9701/11/M/J/22/Q25**[1]**

Which chemical equation with state symbols correctly represents the standard enthalpy change of combustion of liquid methanol, $\text{CH}_3\text{OH}(\text{l})$?

- [A] $\text{CH}_3\text{OH}(\text{l}) + 1.5 \text{ O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
- [B] $2\text{CH}_3\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{l})$
- [C] $\text{CH}_3\text{OH}(\text{l}) + 1.5 \text{ O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
- [D] $\text{CH}_3\text{OH}(\text{g}) + 1.5 \text{ O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is the official IUPAC and Cambridge syllabus definition of standard enthalpy of formation. Option A defines standard enthalpy of combustion. Option C defines standard enthalpy of neutralisation. Option D defines standard enthalpy of atomisation.	[1]
2	D	Option D is correct. By thermodynamic definition, the standard enthalpy change of formation of any pure chemical element in its standard, most thermodynamically stable physical state at 298 K and 100 kPa is assigned a value of exactly zero ($\Delta H^\circ_f = 0.0$ kJ/mol). Ozone (O_3) is an allotrope of higher energy ($\Delta H^\circ_f = +142.7$ kJ/mol). Compounds (H_2O , CO) have non-zero formation enthalpies.	[1]
3	A	Option A is correct. The standard enthalpy of combustion requires: (1) exactly ONE mole of the fuel in its standard state (ethanol is a liquid at 298 K, so $C_2H_5OH(l)$), (2) reacted completely in excess oxygen, and (3) all products in their standard states under standard conditions (water is liquid $H_2O(l)$ at 298 K, not steam $H_2O(g)$).	[1]
4	C	Option C is correct. Strong acids and strong bases are completely ionized in dilute aqueous solution. In both cases, the spectator ions (Na^+ , K^+ , Cl^- , NO_3^-) do not participate in bonding changes. The actual chemical reaction occurring in both titrations is identical: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$, which releases approximately -57.1 kJ per mole of water formed.	[1]
5	A	Option A is correct. Ethanoic acid is a weak acid that is only slightly dissociated in water (< 2%). As OH^- ions consume H^+ ions, undissociated CH_3COOH molecules must continuously undergo ionization ($CH_3COOH \rightarrow CH_3COO^- + H^+$). Breaking the O-H bond to ionise weak acid molecules is an endothermic process ($+\Delta H_{ion}$). This energy requirement subtracts from the exothermic enthalpy of water formation (-57.1 kJ/mol), making the net enthalpy of neutralisation less negative (-55.2 kJ/mol).	[1]
6	D	Option D is correct. Total volume of reacting solution = $50.0 + 50.0 = 100.0$ cm ³ . Mass of solution $m = \text{volume} \times \text{density} = 100.0 \text{ cm}^3 \times 1.00 \text{ g cm}^{-3} = 100.0$ g. Thermal energy released: $q = m \times c \times \Delta T = 100.0 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ K}^{-1} \times 6.5 \text{ K} = 2717 \text{ J} = 2.717 \text{ kJ}$.	[1]
7	B	Option B is correct. Moles of H^+ = moles of OH^- = concentration $\times$ volume = $1.00 \text{ mol dm}^{-3} \times 0.0500 \text{ dm}^3 = 0.0500$ mol. Moles of H_2O formed = 0.0500 mol. Because temperature increased, the reaction is exothermic, so ΔH carries a negative sign: $\Delta H = -q / n = -2.717 \text{ kJ} / 0.0500 \text{ mol} = -54.34 \text{ kJ mol}^{-1} \rightarrow -54.3 \text{ kJ mol}^{-1}$.	[1]
8	C	Option C is correct. Simple flame calorimetry suffers from major systematic heat losses: (1) radiation and convection of heat to surrounding air instead of being absorbed by the water, (2) incomplete combustion of fuel due to limited air supply inside the chimney (indicated by yellow soot on the copper base, releasing less heat than complete oxidation to CO_2), and (3) evaporation of volatile alcohol from the wick during weighing.	[1]
9	B	Option B is correct. Average bond enthalpy is the mean energy required to break one mole of a particular covalent bond (e.g. C-H or C-C) in the gaseous state, averaged across a range of homologous compounds. Bond breaking is always endothermic ($+\Delta H$).	[1]
10	A	Option A is correct. Bond enthalpies apply strictly to molecules in the gaseous phase. In standard enthalpy of combustion measurements (298 K), ethanol is a liquid and water produced condenses into liquid water. The condensation of water vapor into liquid water is strongly exothermic ($\Delta H_{vap} = -40.7$ kJ/mol for each mole of H_2O), which releases substantial additional thermal energy that is absent in pure gas-phase bond calculations.	[1]
11	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
12	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
13	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
14	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
15	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
16	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
17	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]
18	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{products} - H_{reactants} < 0$ (negative).	[1]

25

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
39	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
40	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
41	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
42	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
43	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
44	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
45	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
46	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
47	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
48	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
49	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
50	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
51	D	Option D is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
52	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
53	A	Option A is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
54	C	Option C is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
55	B	Option B is correct. In an exothermic reaction, the chemical potential energy of the products is lower than that of the reactants. Chemical energy is converted into thermal kinetic energy, causing heat to be transferred to the surroundings and resulting in a temperature increase. By convention, $\Delta H = H_{\text{products}} - H_{\text{reactants}} < 0$ (negative).	[1]
56	D	Option D is the formal statement of Hess's Law, which is an application of the First Law of Thermodynamics (Law of Conservation of Energy) to chemical systems: enthalpy is a state function, so the net change depends solely on initial and final states, not on the intermediate path.	[1]
57	C	Option C is correct. By constructing a Hess's cycle with the constituent elements in their standard states at the bottom, the arrows of formation point UPWARDS to both reactants and products. Going against the reactant formation arrows ($-\Sigma \Delta H^{\circ}_f(\text{reactants})$) and with the product formation arrows ($+\Sigma \Delta H^{\circ}_f(\text{products})$) yields: $\Delta H^{\circ}_r = \Sigma \Delta H^{\circ}_f(\text{products}) - \Sigma \Delta H^{\circ}_f(\text{reactants})$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
58	A	Option A is correct. In a combustion cycle, both reactants ($3\text{C} + 4\text{H}_2$) and products (C_3H_8) burn completely in excess oxygen to form the identical combustion products ($3\text{CO}_2 + 4\text{H}_2\text{O}$) at the bottom of the cycle. Combustion arrows point DOWNWARDS from reactants and products to combustion products. Applying Hess's law: $\Delta H^\circ_f = \Sigma \Delta H^\circ_c(\text{reactants}) - \Sigma \Delta H^\circ_c(\text{products})$.	[1]
59	D	Option D is correct. Reaction: $\text{C}(\text{graphite}) + 2\text{H}_2(\text{g}) \rightarrow \text{CH}_4(\text{g})$. Using $\Delta H^\circ_f = \Sigma \Delta H^\circ_c(\text{reactants}) - \Sigma \Delta H^\circ_c(\text{products})$: $\Delta H^\circ_f = [\Delta H^\circ_c(\text{C}) + 2 \times \Delta H^\circ_c(\text{H}_2)] - \Delta H^\circ_c(\text{CH}_4) = [-394 + 2(-286)] - (-891) = [-394 - 572] + 891 = -966 + 891 = -75 \text{ kJ mol}^{-1}$.	[1]
60	B	Option B is correct. $\Delta H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$. Bonds broken: 1 $\text{N}\equiv\text{N}$ triple bond + 3 $\text{H}-\text{H}$ single bonds = $945 + 3(436) = 945 + 1308 = +2253 \text{ kJ mol}^{-1}$. Bonds formed: 2 molecules of NH_3 each contain 3 $\text{N}-\text{H}$ bonds = 6 $\text{N}-\text{H}$ bonds = $6 \times 391 = 2346 \text{ kJ mol}^{-1}$. $\Delta H = +2253 - 2346 = -93 \text{ kJ mol}^{-1}$.	[1]
61	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
62	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
63	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
64	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
65	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
66	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
67	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
68	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
69	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
70	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
71	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
72	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
73	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
74	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
75	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
76	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
77	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
78	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
79	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
80	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
81	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
82	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
83	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
84	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
85	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
86	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
87	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
88	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
89	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
90	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
91	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
92	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
93	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
94	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
95	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
96	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
97	C	Option C is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
98	A	Option A is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
99	D	Option D is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
100	B	Option B is correct. According to Hess's Law, because enthalpy is a state function, the net enthalpy change between identical initial and final states is identical regardless of the route taken: $\Delta H(\text{direct}) = \Sigma \Delta H(\text{indirect steps})$.	[1]
101	C	Option C is correct. Standard enthalpy of formation requires: (1) exactly 1 mole of product in its standard state ($\text{C}_2\text{H}_5\text{OH}(\text{l})$), (2) formed from its constituent elements in their standard, most stable states at 298 K and 100 kPa (carbon as solid graphite C(graphite), hydrogen as diatomic gas $\text{H}_2(\text{g})$, and oxygen as diatomic gas $\text{O}_2(\text{g})$). Diamond is not the standard state of carbon. Option A uses gaseous atoms (atomisation). Option B is the reverse of combustion.	[1]
102	A	Option A is correct. Strong acids (e.g. HCl , HNO_3 , HClO_4) and strong bases (e.g. NaOH , KOH) are 100% dissociated into ions in dilute aqueous solution. The counterions (Na^+ , K^+ , Cl^- , NO_3^-) do not participate in any bonding changes (spectator ions). In all cases, the only chemical process occurring is the exothermic combination of aqueous hydrogen ions and hydroxide ions to form 1 mole of liquid water: $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$, for which $\Delta H = -57.1 \text{ kJ mol}^{-1}$.	[1]
103	D	Option D is correct. $\Delta H^\circ_r = \Sigma \Delta H^\circ_f(\text{products}) - \Sigma \Delta H^\circ_f(\text{reactants})$. Products: $[2 \times \Delta H^\circ_f(\text{Fe}) + 3 \times \Delta H^\circ_f(\text{CO}_2)] = [2(0.0) + 3(-393.5)] = -1180.5 \text{ kJ mol}^{-1}$. Reactants: $[\Delta H^\circ_f(\text{Fe}_2\text{O}_3) + 3 \times \Delta H^\circ_f(\text{CO})] = [-824.2 + 3(-110.5)] = -824.2 - 331.5 = -1155.7 \text{ kJ mol}^{-1}$. $\Delta H^\circ_r = -1180.5 - (-1155.7) = -1180.5 + 1155.7 = -24.8 \text{ kJ mol}^{-1}$.	[1]
104	B	Option B is correct. Using Hess's law for combustion cycles: $\Delta H^\circ_r = \Sigma \Delta H^\circ_c(\text{reactants}) - \Sigma \Delta H^\circ_c(\text{products})$. $\Delta H^\circ_r = [\Delta H^\circ_c(\text{C}_2\text{H}_4) + \Delta H^\circ_c(\text{H}_2)] - \Delta H^\circ_c(\text{C}_2\text{H}_6) = [-1411 + (-286)] - (-1560) = -1697 + 1560 = -137 \text{ kJ mol}^{-1}$.	[1]
105	A	Option A is correct. $\Delta H = \Sigma(\text{bonds broken}) - \Sigma(\text{bonds formed})$. Bonds broken: $4 \text{ C-H bonds} + 2 \text{ O=O double bonds} = 4(413) + 2(498) = 1652 + 996 = +2648 \text{ kJ mol}^{-1}$. Bonds formed: $2 \text{ C=O double bonds} + 4 \text{ O-H single bonds} = 2(805) + 4(464) = 1610 + 1856 = 3466 \text{ kJ mol}^{-1}$. $\Delta H = +2648 - 3466 = -818 \text{ kJ mol}^{-1} \rightarrow \text{approx } -814 \text{ kJ mol}^{-1}$.	[1]
106	C	Option C is correct. Heat released to water: $q = m \times c \times \Delta T = 100.0 \text{ g} \times 4.18 \text{ J g}^{-1} \text{ K}^{-1} \times 9.5 \text{ K} = 3971 \text{ J} = 3.971 \text{ kJ}$. Moles of NaOH dissolved = $\text{mass} / \text{Mr} = 4.00 \text{ g} / 40.0 \text{ g mol}^{-1} = 0.100 \text{ mol}$. Because the temperature rose, dissolution is exothermic (ΔH is negative): $\Delta H_{\text{soln}} = -q / n = -3.971 \text{ kJ} / 0.100 \text{ mol} = -39.71 \text{ kJ mol}^{-1} \rightarrow -39.7 \text{ kJ mol}^{-1}$.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
107	B	Option B is correct. The primary source of error in simple flame calorimetry is heat loss to the cold surrounding air by convection and radiation. Using draught shields prevents drafts from blowing heat away from the copper base, and a lid prevents evaporative and convective cooling of the heated water, drastically improving experimental accuracy.	[1]
108	D	Option D is correct. Average bond enthalpies (e.g. C-H = 413 kJ/mol) are statistical means obtained from many different molecules (methane, ethane, benzene, etc.); the exact bond strength in any specific compound depends on its unique chemical environment. Furthermore, bond enthalpies apply strictly to gaseous molecules; they do not account for intermolecular forces or enthalpies of vaporization/condensation for substances in liquid or solid standard states.	[1]
109	A	Option A is correct. In an energy level profile diagram, the products lie 150 kJ/mol lower than the reactants. To go from products to the transition state (reverse reaction), one must climb: (1) the energy difference from products back up to reactants (+150 kJ/mol), PLUS (2) the forward activation energy from reactants to the transition state (+60 kJ/mol). Thus $E_a(\text{rev}) = 150 + 60 = +210 \text{ kJ mol}^{-1}$.	[1]
110	C	Option C is correct. The standard enthalpy change of combustion, ΔH°_c , is defined for the complete combustion of exactly ONE mole of substance in its standard state (methanol is liquid $\text{CH}_3\text{OH}(\text{l})$ at 298 K, 100 kPa) with excess oxygen gas ($\text{O}_2(\text{g})$). Under standard conditions (298 K), carbon dioxide is a gas ($\text{CO}_2(\text{g})$) and water is a liquid ($\text{H}_2\text{O}(\text{l})$).	[1]